Combining XR and AI for Integrating the Best Pedagogical Approach to Providing Feedback in Surgical Medical Distance Education

Yves S. Nkulu-Ily(✉)

University of Kentucky, Lexington, KY 40506, USA
lyshabo@hotmail.com

Abstract. This position paper maintains that learners' involvement and accurate representation through cues and stimuli are essential aspects of an AI-guided XR app for learning surgery. It then shows how and why it is crucial to conduct expert interviews when designing/developing such an app. A multidisciplinary team of 22 experts was interviewed to provide insights into new visual technologies' intercommunication (i.e., Extended Reality-XR, Artificial Intelligence -AI or Machine Learning-ML, and 360° videos) and the pedagogical aspects (i.e., educational theories, techniques, and practices) of designing an XR system supported by an ML-based subsystem. These insights are about instructional design elements to maximize the impact of simulation-based training (SBT). The paper guides educational games for medical surgery design and user experience theory and concepts. It shows how to capture the actual task's and environment's fundamental characteristics expressing realistic behaviors. It shows the significant issues (e.g., occlusion and the use of instruments) in capturing these tasks and possible options to overcome them (e.g., using video feeds from endoscopies and robotic surgery). It also provides practical opportunities for facilities in resource-constrained regions lacking internet access/reliability/speed. For qualifying impact, this paper provides the best way to measure learning experience return on investment for medical education.

Keywords: extended reality · artificial intelligence · intelligent system

1 Introduction

The higher level of realism in implementing appropriate pedagogical theories, practices, and techniques for medical education, the better the learning process is continuously modified during the performance and even post-hoc. At least, that is what is expected of orthopedic physicians training in an extended reality (XR)/artificial intelligence (AI) and distance education (DE) environment. I.e., the media (technology) cannot influence learning. Learning is influenced by pedagogical methods used in the media (or by a live teacher with the same amount and type of learning).

Making things hard is the trying to incorporate the right way to learn better, and desirable difficulties of Bjork best demonstrate how surface learning often fails in the

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long run. Desirable Difficulties or a considerable but desirable amount of effort in the initial learning period is a research-based principle for teaching and learning, leading to improved long-term performance [5, 6]. This difficulty level should involve a small investment of time and effort in spaced activities, calling for generation and self-testing through feedback to translate into significant learning gains identified as a desirable level of difficulty opening a window of learning opportunity [20]. It also looks like the information processing theory and the descriptive theory of skill acquisition align with the Challenge Point Framework (CPF) [21] and desirable difficulties [9]. I.e., challenges in a learning environment should match learners' abilities. The CPF, coined by Guadagnoli and Lee [14], helps in the effects of various practice conditions' conceptualization in motor learning by relating practice variables to the individual's skill level, task difficulty, and information theory concepts. Hence, any raises in task difficulty may considerably impact learning potential with a reverse impact on performance, i.e., maximizing learning while minimizing any harm to performance in practice determines the optimal challenge point (OCP).

It is believed that such a learning environment can create a model of success in which learning becomes inherently rewarding and enjoyable. However, "students may not engage in retrieval practice in their natural educational settings due to lacking metacognitive awareness of the mnemonic benefits of testing." Retrieval information practice through testing the latter positively impacts long-term retention and learning [19, 25].

In different terminology, Metcalfe's [20] Region of Proximal Learning, Hitchcock and Mcallister's [17] OCP, and both Hartwig and Dunlosky's [16] Zone of Proximal Development (ZPD) and Desirable Difficulties refer to a similar concept, that students engage in independent learning and practice, which should be pretty tricky, but at an acceptable level. Students are challenged enough to maintain their focus, and they can learn new concepts with guided assistance and scaffolding. Then, as the learning proceeds, the support structure is slowly removed. While Verenikina [30] refers to the ZPD as scaffolding's theoretical foundations, De Guerrero and Villamil [15] refer to scaffolding as ZPD's metaphor. ZPD and scaffolding also seem synonymous in that they both suggest learning with adult guidance or more capable peers' collaboration exceeds independent learning. The ZPD, or OCP, is a tool instructor can use to make complex decisions about when to give extra scaffolding.

Reiser [23, 24] points out that scaffolding is possible by structuring and problematizing and argues the interest in taking advantage of software tools to scaffold learners' complex tasks. These software tools can be used in structuring learning tasks while guiding learners through critical elements that support their planning and performance.

Since learning strategies creating desirable difficulties in the learning environment are most effective for learning outcomes' long run, Biwer et al. [4] demonstrated that these findings are reliable for practical students' training in higher education (i.e., positive effects on knowledge about effective learning strategies and increased practice testing use). Finally, beyond OCPs' potential maximizing for surgical training, Gofton and Regehr [13] also stress the irreplaceable value of a mentor.

These approaches – i.e., OCP, desirable difficulties, scaffolding, and ZPD – seem related and should not be considered separately for their application in a medical or surgical AI/XR environment where practicing is a must. Such an environment has a high cognitive load (CL) for learners unfamiliar with it. Software tools can be leveraged for that purpose, and the interactivity effect of CL theory elements can help determine when to achieve "Desirable Difficulties." Hence, this paper aims to investigate how to integrate pedagogical aspects by combining AI, XR, and 360° videos to improve the learning curve in Orthopedic Surgery DE.

2 Background/Purpose

This study will explore the use of XR, AI, and 360° videos in orthopedic surgery DE, focusing on understanding the appropriate level of realism for different learning stages (1–5 categorization) and using deep learning (DL) algorithms for skill training. An XR/AI system for competency-based training and assessment in orthopedic surgery DE, with live instructors lead 360° Procedure training with prerecorded elements, programmatic skills modules using augmented reality (AR), and AI-driven models for visual performance assessment, will be developed.

As a potential further extension to the subsystem, a rule-based agent (i.e., CPF), with either tutor-specified or automatically learned rules, will be used to implement an 'Adaptive Course Sequencing System (ACSS),' such as Alzahrani et al. [2] model. Depending on students' pedagogical capabilities and immediate needs, their intelligent pedagogical tutoring system can autonomously adjust its level of guidance; the instructor or students can do so too. The to-be-implemented ACSS will allow residents to grant the subsystem a level of autonomy in guiding them through cases of varying levels of difficulty and on different issues to assist them in becoming sufficiently competent in one subject before moving to another and suggesting appropriate sequences of points to optimize each resident's learning experience.

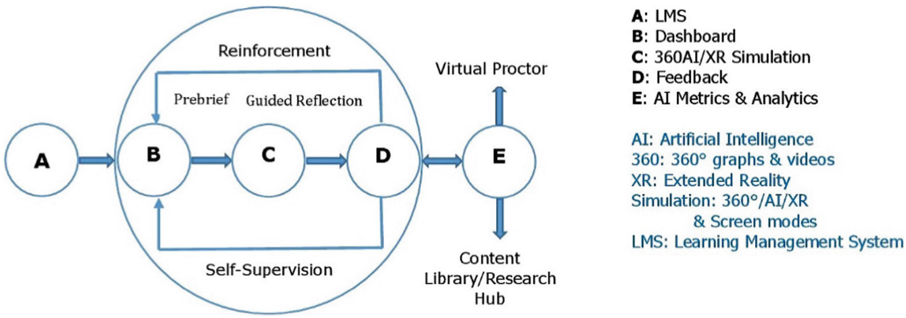

Fig. 1. Learning Model

The focus is also on understanding how to use AI to provide feedback for reinforcement and assessment within the XR application and effectively communicate that feedback to users (see Fig. 1). In the trial, the intervention will be feedback to surgeons/medical students; 50% get feedback from technology and the other 50% from a human instructor.

3 Materials/Method

Interviews and user surveys, with follow-up interviews, were conducted via email or Zoom, and 22 pieces of feedback from experts from the industry and academia were gathered. A few interviewees provided detailed, discipline-centered answers (i.e., sims in surgery and medical education). Some also did not have the time or inclination to write/answer the immediate Ph.D. length response needed to answer these questions correctly. Some others probably gave helpful answers to about 25% of them to prioritize their current deadlines and paid work.

Based on the questions (see Appendices), user experience (UX) departments were contacted, especially for answers to human–computer interaction (HCI) questions related to AI and XR applied to surgical training. The other lines of inquiry touched on quite a few topics, including the application of AI (1) for pedagogical purposes, (2) for tracking a candidate's movements, procedures, etc., based on video observations, and (3) for assessment and feedback.

Twelve questions were more technical and back-end related, so data science or computer science experts were contacted for their perspectives. Nine out of the seventeen questions focused on medical and pedagogical aspects of the simulation, so experts from Educational Technology, Medical Education, and Orthopedic Surgery were sought for their perspective. Finally, industry experts and an international research organization were reached out to for synergetic expertise. (see Table 1).

Table 1. Summary of the number of experts per category.

Academia			**International research organization** (AI)	**Industry** (e.g., AI, XR, Robotic surgery, Games, etc.)
User Experience/UX	**Data Science or Computer Science**	**Educational Technology, Sims in Surgery, Medical Education, & Ortho Surgery**		
3	9	6	1	3

The need to use semi-structured user surveys/interviews with foundational questions was to propose insights improving XR medical education applications' design knowledge for non-Medical XR people. It was also unclear to which extent the XR application and AI would communicate with one another to integrate pedagogical aspects.

4 Practical Implications

The goal of developing an AI-guided XR tool for learning surgery (Orthopedic surgery) in DE is to understand how learning works in a medical education context. Most aspects of the learning trajectory are certainly similar for both an orthopedic and an appendix surgeon. The overall question is how to ensure the simulator training skills transfer to real life.

Procedural motor learning requires practice in a realistic environment. It also involves variation in the training and adaptability of the training difficulty to match the student's progress and improve both long-term memory and transfer of learning. It is essential for motivation and efficiency of learning to keep challenge and ability in lockstep. Otherwise, people get bored or intimidated by the tasks being too challenging, both of which result in the individual losing motivation to complete the training.

Using the CPF adds variability in guidance that can be helpful to the learner. CPF simulations can be used to provide training examples so that the AI model can be trained. After training the AI model, the simulation may incorporate the model to show how well the AI model would work. Simulation is based on some formulas and assumptions, but actual operations from learners can be any type and in the wilderness/real-world (i.e., they are not controlled, and many unexpected operations/scenarios may occur).

OCPs should be determined for each level of the learner. Each surgical skill or task will have its own learning curve that will be individualized for every learner. A decision should be made about where the limits of simulation are. The setback is that, at some point, the OCPs will require actual patient encounters; learning in the simulation will be asymptotic and a waste of time. Fortunately, deliberate practice, when used in addition to mentorship, is an excellent method to improve learner performance while shortening the growth phase of the learning curve [7, 12].

4.1 How Will the AI Determine a Failure or Successful Event?

In terms of how the AI will determine a failure or successful event, it may be helpful to define specific criteria or metrics that the AI can use to evaluate performance. These criteria or metrics could involve using data on the user's movements and actions within the simulation, as well as other relevant factors such as time taken to complete the task or the accuracy of the user's actions (e.g., through imitation by labeling images/videos and comparing "Ghost images" vs. "Real-world images," [11]). However, generally speaking, AI is only "intelligent" in finding patterns in data. With the right AI algorithm, good training data, and optimal parameters, the AI could effectively be trained to recognize a "failure" or "success" event by seeing enough examples of each event. I.e., DL will need to be trained on hundreds of hours of video labeled by experts. Good refers to quality, right relates to functionality, and optimal refers to a balance between different attributes, such as performance vs. safety.

The alternative would be that, rather than using a proper "AI," the designer/programmer could manually determine a set of rules or heuristics that a program can use to determine failure or success. AI systems are generally based on goal functions (also called utility functions or the results from "criteria or metrics"). To know how to react (no matter if the AI learns or reacts), the AI system must compare alternative "actions" or picks to make decisions. Even in some expert systems, where no goal functions seem to be implemented, there are goal functions in place, but they are binary (generate true or false values for each step). Most of the work could go into the design of such a function. Most code in a chess computer player is based on the chess rules, which is the foundation of how the goal function can assess each "game position."

The AI could then use this information to provide feedback to the user and assess their performance. It may also be helpful to consider how the AI can be integrated into the XR application in a way that is seamless and intuitive for users. Whether the trained AI evaluator eventually is to be built into the XR application would reduce friction (i.e., it would allow for a better user experience). Yes, integration could be advantageous if program complexity can be held at a minimal level. There is no need to integrate it into the XR application as long as it is connected to the internet. However, once the model is trained, it should be relatively light on resources, so building it in should be possible.

4.2 AI Integration into the XR Application

For the integration of the AI into the XR application, the considerations should weigh the AI's performance needs/power consumption, especially when running on low power or low battery capacities devices. Mobile phones/tablets or standalone Head-mounted displays (HMDs), and response times – both from the perspective of how fast the AI can compute on the target hardware, as well as any latency introduced by having the AI running off device (e.g., in the cloud). For low-resource countries, another consideration is internet access/reliability/speed. Though running this AI off-device/in the cloud (i.e., content library; see Fig. 1) in places where internet access is reliable and fast enough would not be advantageous, it would serve as a backup solution in low-resource countries.

Also, any hardware integration has its challenges. Using any XR with less than 60 Hz frame rate is not advisable because users will get nauseous. For the other applications, this determination will be based on the HCI and how close it is to the natural motion to which the skill will transfer. Holding a mouse or using a keyboard is a far-transfer skill for many of the skills taught unless they are cognitive or knowledge-based skills.

Why is XR HCI a critical consideration too? XR HCI is not natural to the inputs used in surgery. Although some surgical robots use Xbox-type controllers (Auris), most do not have any joystick similar to what real surgeons use. XR applications that use natural hand motion have not had excellent fidelity for inputs and again have the same issue with occlusion and instrument handling. Some folks are looking at 3D-printed controls that look similar to the real thing but have built-in sensors (accelerometer, gyroscopes, etc.). Many commercial entities in the orthopedic space using XR (e.g., OssoVR, Precision OS, etc.) have some data, but they have not yet been validated or transferred to surgical skill. XR has a high CL for learners unfamiliar with it and can actually detract from the learning.

How will the AI be integrated into the XR application to provide feedback? The guidance AI needs for the kind of feedback that will be provided is in terms of the values of the indicators established for data format or transfer process (i.e., the scales users are evaluated on). What is the point here? The guidance AI needs should be related to the type of categorization (e.g., Likert scale 1–5 or 1–10, etc.) for different learning stages.

The Likert scale is a simplification to allow for expert review. 1–10 does not allow for humans to really make a sound judgment and drops the inter and intra-rater reliability. AI does not have this problem. It will analyze the experts (with a tight, narrow definition) to create the algorithm. The economy of motion has been the most significant area researched that can distinguish between novices and experts. That is on a continuous variable and not Likert.

Given enough high-quality data and training time, a DL model should be able to learn an association between a video of a novice and a video of an expert. The question is really whether those ratings are satisfactory. With a clear definition of success states for each scale/level, the experience levels can be gated; i.e., users must succeed in the first level (or scale) to go to the next one. In each part of the experience, a decision must be made on what constitutes success and if the user can repeat the level to achieve a higher score.

Users will be evaluated on a scale of 5 (skilled) to 1 (novice). However, scaling is dynamic according to needs, so any scaling method can be used depending on the AI system design. The same scale or different scales can be used. An attractive alternative could be to use raw data as input for the AI machinery to estimate probabilistic states [28]. A continuous variable would be used instead of a Likert one. However, if different scales are used, they must be rescaled to keep values consistent, i.e., the AI scale must be mapped with the one used in XR.

Realism and Sophisticated Behaviors

Based on a Likert scale measurement, how much fidelity would be needed? The question of how much fidelity is needed is hard to answer because the measurement of performance or learning is pretty blunt (1–5 categorization). Many things are being learned: anatomy, clinical knowledge, motor skills, procedures, and problem-solving. These can all be measured more precisely than 1–5 categorization. Each stage has specific goals (say, instrument familiarization and location in the training device).

Moreover, there is no quick fix to design for anyone to capture and analyze sophisticated behaviors in real time for skill training. Learning is influenced by teaching methods provided in the media or by a live teacher with the same amount and type of learning. Teaching is complex, though, and no 'best practice' can be transferred from one situation to another. Expert teachers intuitively assess each learner's current cognitive knowledge and affective state through engagement with the learner. How will this complexity be addressed? The app, through the ACSS, will automatically use the ZPD or OCP to make complex decisions about when to give extra scaffolding.

Given that teaching is complex, what useful strategies could then be used to design the application to capture and analyze sophisticated behaviors in real-time? To design the application to capture and analyze sophisticated behaviors in real-time, high-speed videography and motion capture technology may be considered to replicate a tactile experience with real haptic feedback.

The preceding reasoning could also involve using 3D assets and hardware that provide haptic feedback to the user, such as haptic gloves or other wearable devices. However, haptic feedback devices are very cool. Unfortunately, this technology is still in its infancy to date, and therefore, there are not yet any well-supported, commercially available haptic feedback devices that would work for this kind of app.

Moreover, AI models cannot currently learn these sophisticated and tacit processes. The only way forward is to develop AI-human interfaces that help teachers and/or learners with these complex interactions. AI will catch the behavior of the user and see if the behavior is at or close to the right move. DL part will also require a lot of labeled data with many example videos (perhaps hundreds of hours) labeled by experts.

There are commercial entities actually working on this aspect. With computer vision, movements can be tracked; the most significant issues obviously come with occlusion and the use of instruments. Some companies are working on overcoming these. The most successful applications (like Proximie) use video feeds from endoscopies and robotic surgery where only instruments are tracked, not the hands. The preceding is a precious line of research, in any case. To allow anyone to capture, equipment and setting must be standardized. This is why laparoscopy and robotics are the first areas to look at this.

An attractive alternative would be to add educational games with stealth assessment and built-in learning dashboards for instructors to provide real-time feedback [27]. The continuous interaction with stealth assessment produces rich sequences of actions as data points that the stealth assessment captures in log files. Stealth assessment rubrics automatically score these aggregated data in real-time by Bayesian networks (or other statistical models). This process shows evolving mastery levels on targeted competencies [27].

4.3 AI-Human Interfaces or Educational Games' Best Pedagogical Practices

In determining the best pedagogical practices for these AI-human interfaces or educational games, the best practice is scaffolding the learner depending on their abilities and performance. The best application will keep them engaged at the edge of their abilities and increase the difficulty over time. The best formative will give them subtle feedback, and the experience guide them in what is right and wrong (similar to video games). Real-time feedback has to be immediate, succinct, actionable, and straightforward. Education around performing the task should be embedded in the experience. Summative performance data should delineate their skill level in separate domains (e.g., accuracy, efficiency, etc.).

In learning applications, pedagogical agents are usually used. I.e., some characters communicate with each other with the user. This agent will also provide feedback later. In some applications, for example, a teacher or scientist character guides the user. AI needs to learn from each domain of the skill the novice's delta from an expert. For example, they may be accurate but not efficient. This information needs to be communicated in some way to both the AI to evaluate and feedback to the user.

In terms of using AI to provide feedback for reinforcement and assessment, it would also be essential to consider what information is most relevant and valuable for the users to receive. It may also be that showing/not showing this information is more appropriate/valuable at different stages in the educational pipeline. This decision may depend on the stage of learning and the specific skills being trained [26]. However, formative feedback is best served immediately and summative after each experience or set of experiences.

For example, novice learners may benefit more from detailed explanations of how to perform a task, while more experienced learners may benefit more from feedback on their specific performance. It may also be helpful to consider different ways of communicating feedback to users, such as through visual or auditory cues within the simulation or more detailed reports or analyses provided later. Feedback should be multimodal (sounds, lights, vibrations, etc.) and nearly immediate. There are UX guidebooks on best practices. It should not distract from the experience but be very clear and succinct to the user, as in video games.

At first, it would be desirable to try to make it accessible later. If it is built within the simulation, then only with sufficient data can the AI model tell if anything is wrong on the fly – the goal is to correct behaviors in real-time. Likely all the information listed will be needed, which means the context is leaning towards game design and user experience theory and concepts here—like a car racing simulator or Game, such as Forza 7.

Regarding the relevance of the Game's use, a few points must be considered carefully. Though serious games for medical education and surgical skills training can apply here, there are several concepts related to using games in medical learning. These concepts include:

1. The use of games in medical practice is still limited, and their use is only limited as a companion to the leading learning media. The understanding is that only AR and Virtual Reality (VR) expose medical models, and only a few have interactions (such as surgical practice).
2. Referring to the articles on JMIR Serious Games, the use of games is still limited to understanding the material, not the practice of surgical operations.
3. There are several VR applications with educational content on human anatomy, but there are still very few specifically related to surgery.
4. There are some medical games, but they are only for fun, not entirely educational games.
5. The experience must be embedded in an environment that reinforces skills through the gameplay without blindly putting on badges, scoring, and leaderboards. These should be meaningful to the experience.
6. Game-based stealth assessment seems valid, reliable, and improves learning [27].
7. For DL, medical expert systems are suitable, and Ripple-Down Rules (RDR) may be more relevant [10]. I.e., combining ML and RDR, particularly RDR systems that recognize a case outside their competence. RDR is easier to implement on insufficient data than AI.

4.4 A Pedagogical Approach to Errors in a Simulated Environment

Educators teach students about errors to help avoid them. Residents can make mistakes in the simulated environment since that is how they learn, so they should fail in a simulated environment rather than in real life. However, in general, the understanding of the research is that teaching about errors tends to increase the mistakes, not decrease them. In medicine, many (some say most) errors are due to missed condition/action situations. For example, forgetting to do something, and the second most frequent error is "doing something wrong," missing steps in a procedure, or implementing the wrong approach for the conditions. In both cases, the error is sometimes due to the instructor skipping over a step or steps in a procedure/assessment that misses the student's failure to remember the actions or process. Adequate instruction, practice, and corrective feedback should help avoid future errors. In some cases of coaxers, students become distracted and cause errors despite solid education.

4.5 Multi-device Considerations When Designing/Developing the App

When developing an app. to be used for multi-devices, then one consideration that should be taken into account is whether this app can be available for multiple devices or require XR devices. The only natural way to understand it is to actually trial it since these devices have many pros and cons (See Appendices, Table 2). However, potential issues with any preferences can be factored in at the start of a process. I.e., eliminating specific approaches due to technical limitations, pedagogic goal matching, budget constraints, and assessing possibilities.

Here again, the cost is the most significant consideration. Also, any hardware integration has its challenges. Using any XR with less than 60 Hz frame rate is not advisable because users will get nauseous. For the other applications, this determination will be based on the HCI and how close it is to the natural motion to which the skill will transfer. Holding a mouse or using a keyboard is a far-transfer skill for many of the skills taught unless they are cognitive or knowledge-based skills.

Broadly processing/computational power differs across all the devices, and the delivery modality (i.e., User Interface or UI and UX) will (or at least should) differ depending on whether mobile, XR headset, desktop, etc. The Oculus Rift is a high-powered 3D headset, whereas a Google Cardboard uses phones, so its power will depend on the quality of the user's phone. VR headsets can provide 3D video, whereas smartphones, tablets, etc., will not.

Fortunately, real-time engines (e.g., game engines Unity or Unreal) provide XR, support multiple platforms, and can connect to MATLAB by socket connection. So, an educational experience can be deployed to multiple platforms if the app is developed correctly to ensure it runs on multiple platforms.

4.6 Data on Decisions and Actions Source

Event data is excellent depending on how the analytics are hooked up moment by moment. Still, a data plan is needed to clean, analyze, and visualize meaningful data. Secondary validation information like the Objective Structured Assessment of Technical Skill (OSAT) and expert reviews can help validate the AI assessments. I.e., the outputs will need to be compared to some established, validated form of data. XR and AI materials are all predicated on what is being done.

Moreover, AI is good at classification and regression. Generation based on such technologies as stable diffusion is also in fashion. However, it is bad at identifying what is not known as unknown. It even tells seemingly plausible lies. Simply, it is lax, which is AI's limitation today. When 100% guaranteed information is a requirement, it is too harsh for AI.

Reinforcement learning algorithms are probably appropriate when inputting users' scales to train AI systems is required. Getting information on a finer scale would be better, but if it is difficult to require that of humans, it would be advisable to use 1–5 scales.

4.7 Used AI Methodology/Realized Experimentation in Neurosurgery

The Force sensor (i.e., educational system) lays a basis for integrating AI and VR simulation into surgical education with an algorithm distinguishing between two groups of expertise, objective feedback based on proficiency benchmarks, and instructor input [22]. Its rich digital output can establish a robust skill assessment and sharing platform for orthopedic surgical performance and training. Beyond distinguishing between groups of expertise, objective feedback based on proficiency benchmarks, and instructor input, the force sensor (i.e., using the support vector machine learning model) [22], SmartForceps data app [3], and tool-tissue forces in Newton (i.e., using deep neural networks) [1] can help improve training in orthopedic surgery and allow avoidance of these errors and resulting complications. The latest deep learning models (i.e., deep neural networks) can capture more parameters for performance metrics and perform more realistic simulations.

4.8 Learning Experiences in Healthcare Return on Investment (ROI) for Education – Measure

"The level of expertise of surgeons performing procedures should be reported" to determine whether the technique or surgeon's poor execution – of the technique – led to poor results. Not only is comparing results between surgeons with significantly different levels of expertise invalid, but this critical information is also generally missing from published articles. [29] A well-defined expertise definition includes (1) Non-specialist, (2) Specialist – less experienced, (3) Specialist – experienced, (4) Specialist – highly experienced, and (5) Expert [29]. Whether expertise or performance is measured, performance is usually measured by OSATs. However, surgical procedures' performance and improved learning curves are rarely measured [7].

The technology vs. human comparison is a 'teaching method one vs. teaching method 2' study of the kind Clark [8] discouraged many years ago – on the ground, such comparisons are confounded, always. In this case, the question is: how does the feedback differ between the two conditions (other than the delivery via machine vs. human)? If the feedback has the same information, frequency, and structure, there should be no difference in outcomes. If they are different in any of these regards, those differences presumably underlie the results. If, for example, the machine can provide more or more frequent feedback and/or practice, then this is perhaps the basis of the difference, not the circumstance of machine vs. human.

A suggestion to make the subsystem more "intelligent" would be by noting residents' responses to individual cases in a way previously done for students learning computer programming [18]. Their system records every student interaction with it, providing a rich data set for analysis. Then, statistical pattern recognition and ML techniques automatically identify the "common mistakes" for each type of case. This improved subsystem should respond appropriately to a more comprehensive range of resident errors, including the possibility of mistakes that even an experienced tutor may not have anticipated.

5 Conclusion

This qualitative interview is not a general survey, and these are complicated questions that require either substantial experience in this area of research or research investigation in their own right. Many of them are complex and not necessarily "solved" – many are thesis topics on their own that are probably better answered by someone with a background in education (i.e., educational technology or medical education).

This paper describes early-stage research that is setting out to investigate how AI, XR, and 360° videos should communicate to integrate the best surgical medical education approach from the perspective of an Instructional Systems Designer. However, the biggest question in this research is where do learners need to be on the learning curve before and after the XR experience. Can they start at zero (XR is their first experience with that surgery), or do they need some base knowledge? Different interactive elements of XR will support different levels (i.e., a learner just needing to know anatomy does not need a highly interactive, game-based experience to build foundational knowledge, but a more advanced learner will). Ultimately at which point is XR not enough for the learner? When do they need actual cases to advance their learning?

According to Wang [31], no perfect product allows the fusion of AR, AI, and machine translation technologies. So, the intention should not be to fuse the models/technologies. AI and XR may be used together collaboratively without being fused. The RDR can be combined with the CPF and stealth assessment with built-in learning dashboards to help residents avoid mistakes while improving long-term retention and transfer of learning.

Rules should be used to establish WHEN and HOW to deliver feedback to the learner. Different types and timings of feedback are applicable to different users. Finally, a straightforward way to design a goal function is to use the error between the resident's operation and the instructor's operation – because recorded operation steps and trajectories, and those from the residents are already available, they may be compared to get the errors. Other kinds of goal functions may also be used.

Appendices

Table 2. Hardware's pros and cons.

Other Devices				XR Devices	
Smartphones/Tablets:		Desktop/Laptop:		HMDs (e.g., Oculus Rift, HTC Vive, or Google Cardboard):	
Pros:	Cons:	Pros:	Cons:	Pros:	Cons:
• Widely accessible • Most people are familiar with them, • Lots of sensors, and • Interaction methodologies	• Low performance, • Low battery life, • The form factor is not conducive to certain types of interaction (e.g., having to hold the device in one hand while using the other hand to interact)	• Widely accessible, • Most people are familiar with them, • Best performance, battery life	• Fewer sensors and interaction methodologies, • Generally suited for 2D "flat surface" experiences using a keyboard and a mouse	• Immersive, • The best form factor for some experiences (e.g., both hands are free to interact, best support of natural hand interaction)	• Less accessible, • If untethered worst performance and battery life, • A novel technology for most people, which means they will perform worse but still prefer it.

Sample Questions (Interviews/Questionnaire).
The questions are way too detailed and the application domain very specialized. A lot of this set of opening questions are early questions, which means that refinement or more context will be given to answer where possible.

1. How can I design for anyone to capture and analyze sophisticated behaviors in real-time in applying the field of high-speed videography of movement and its analysis by DL algorithms to skill training?
2. Data format or transfer process: Users are evaluated on a scale of 5 (skilled) to 1 (novice). Should that be the same scale used in values sent to the AI?
3. Would it be better for the trained AI evaluator eventually be built into the XR application?
4. What is the best practice regarding pedagogy here? I'm also not sure exactly what information I will have to show the users - is it information on how to perform the task, their performance in the task, or something else?
5. Would it be advisable that this information be shown to the users within the simulation, or would it be better if it is accessible later?
6. How will AI be used in application development to provide feedback for reinforcement and assessment?
7. What is the best pedagogical approach to providing feedback, and how best to communicate that feedback?
8. How will AI determine a failure or successful event since AI needs some guidance for the kind of feedback that will be provided?
9. Which "serious games" for medical education and surgical skills training would apply here, please?
10. What type of data do you think will be helpful for my study, and what collection method would you recommend? Additionally, can you outline the development process for producing XR/AI materials, please?
11. I want the app to be available for Smartphones, Tablet, PC / MAC (desktop/Laptop), moreover, will also require the use of VR devices such as a Head-Mounted Device (HMD) like the Oculus Rift, HTC Vive, or Google Cardboard. What could be the potential issues with my preferences, please?
12. Please outline the development process for producing materials using the Challenge Point Framework (CPF) (graph or diagram).
13. Or can you please share some practical applications of the CPF to simulation-based medical education and training?
14. For example: How can this framework be incorporated into a simulation?
15. What are some of its potential limitations, and how can we address them?
16. What 3D- assets, Motion capture technology, and hardware are there that could be implemented to replicate a tactile experience with real haptic feedback?
17. Using a technology-authoring environment, how can you mock-up or storyboard a prototype for a type of learning experience you may envision learners going through to answer a research question related to AR/VR, please? (i.e., an Example or Sample in any aspect of surgery to give me an idea of such a prototype)

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